Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°4 Réseau II

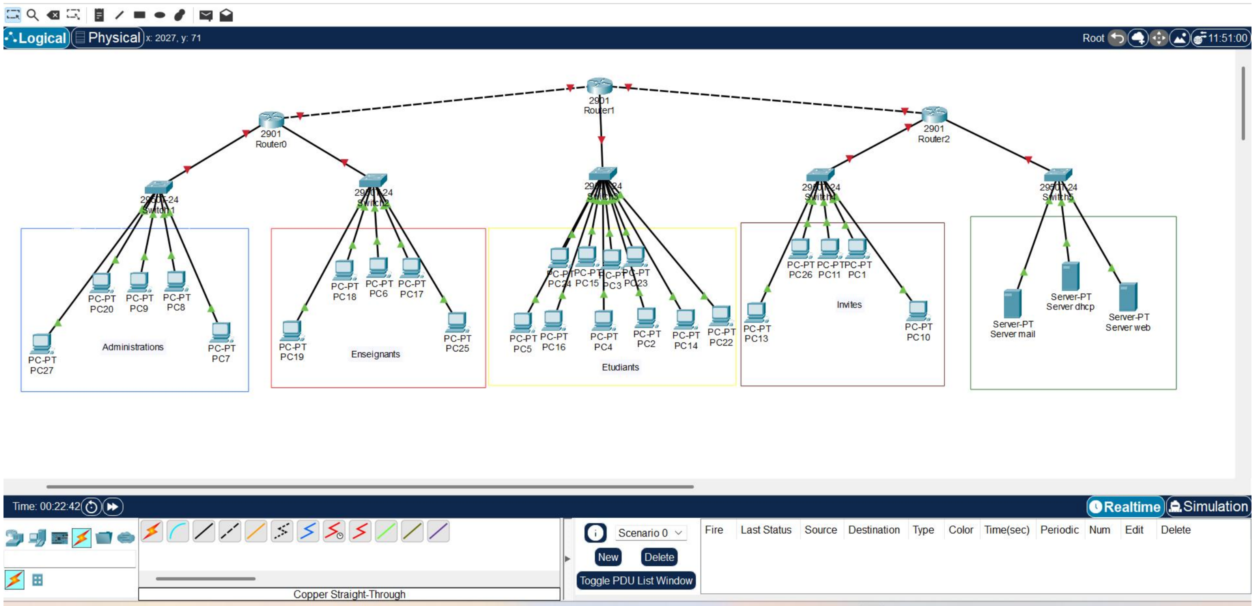
Nom & Prénom : COFFY Cliford

Niveau : L3

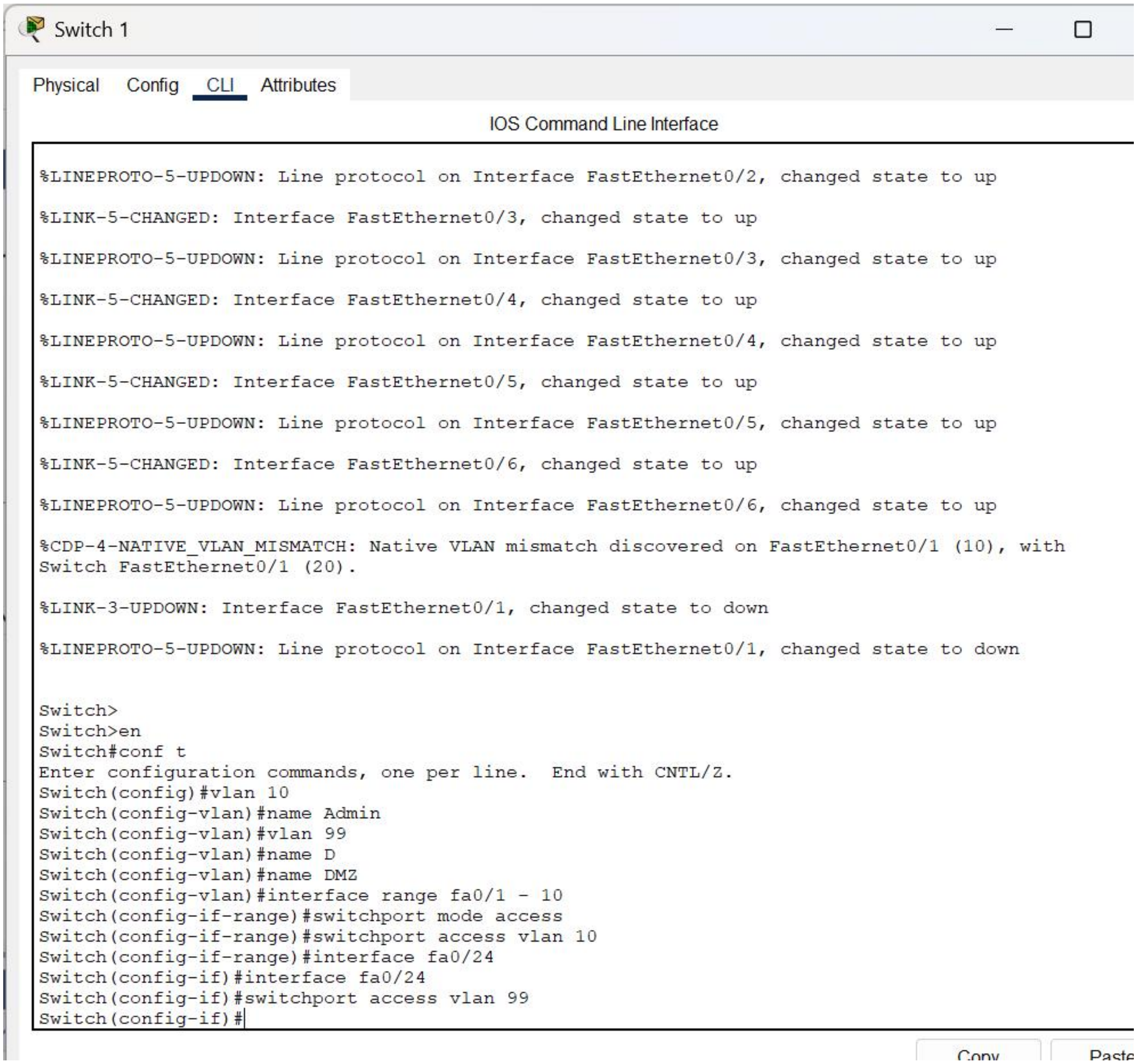
Prof. : Ismael SAINT AMOUR

Date : 18/04/2026

Ici je commence avec l’illustration de mon topologie



Configuration des
switchs



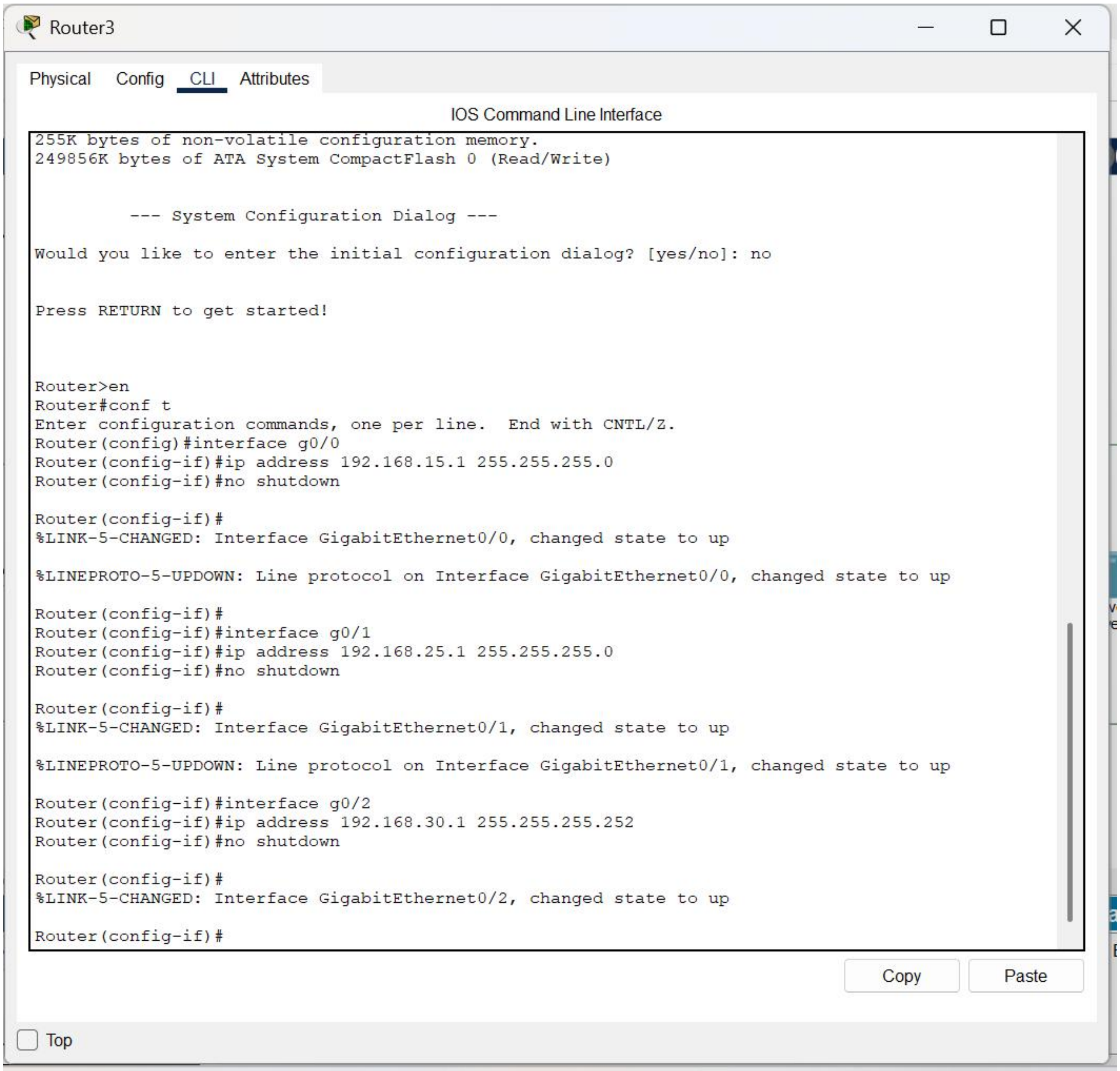
Switch2

Physical Config CLI Attributes

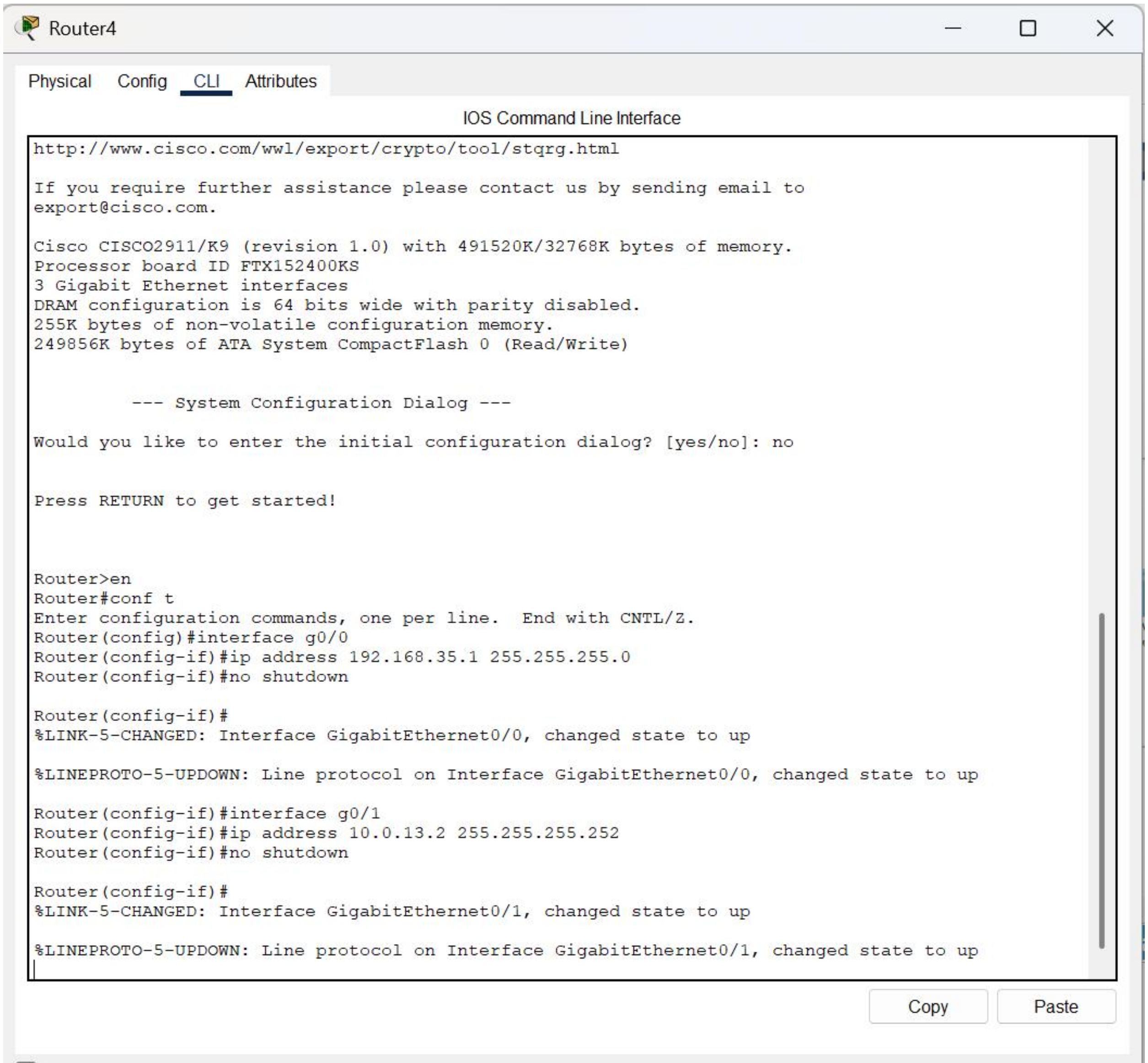
IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (20), with
Switch FastEthernet0/1 (10).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with
Switch FastEthernet0/1 (30).

%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down


Switch>en conf t
          ^
% Invalid input detected at '^' marker.

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name Enseignat
Switch(config-vlan)#name 99
Switch(config-vlan)#name DNZ
Switch(config-vlan)#interface range fa0/1 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#interface range fa0/1 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

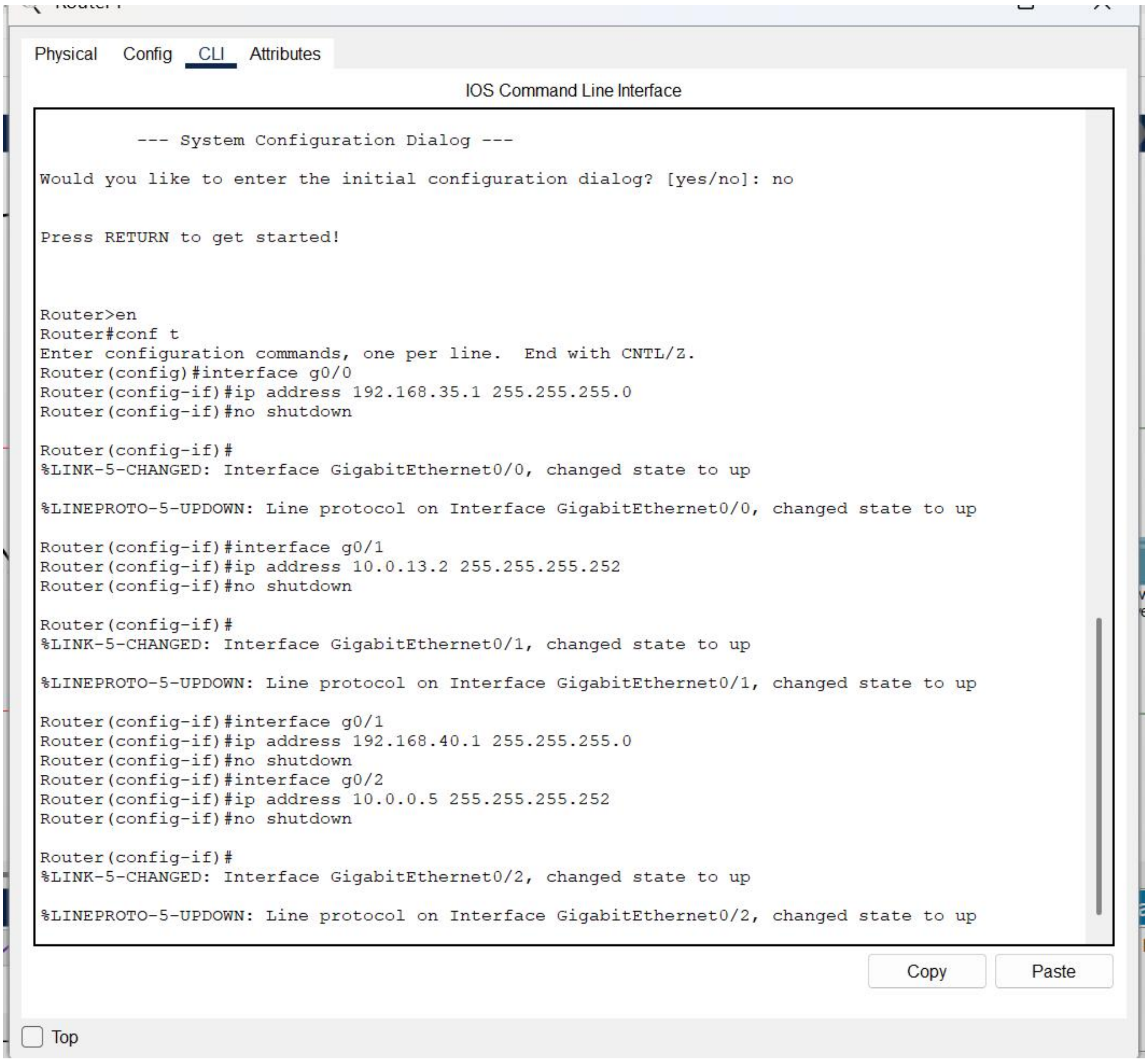
Configuration des 3 Routeurs




```
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.15.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.25.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.30.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
```

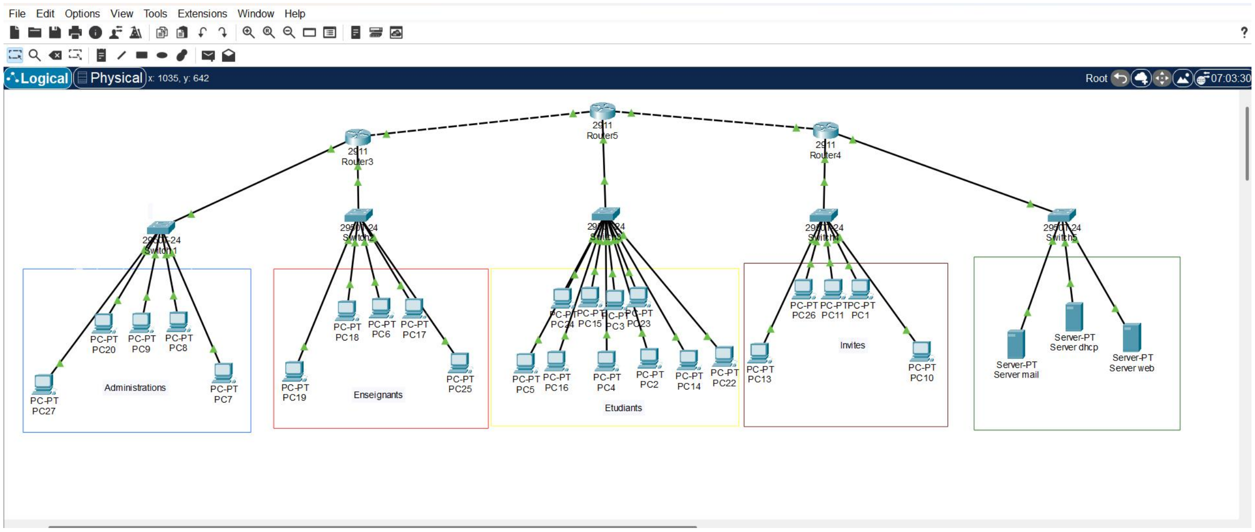



```
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.35.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.13.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

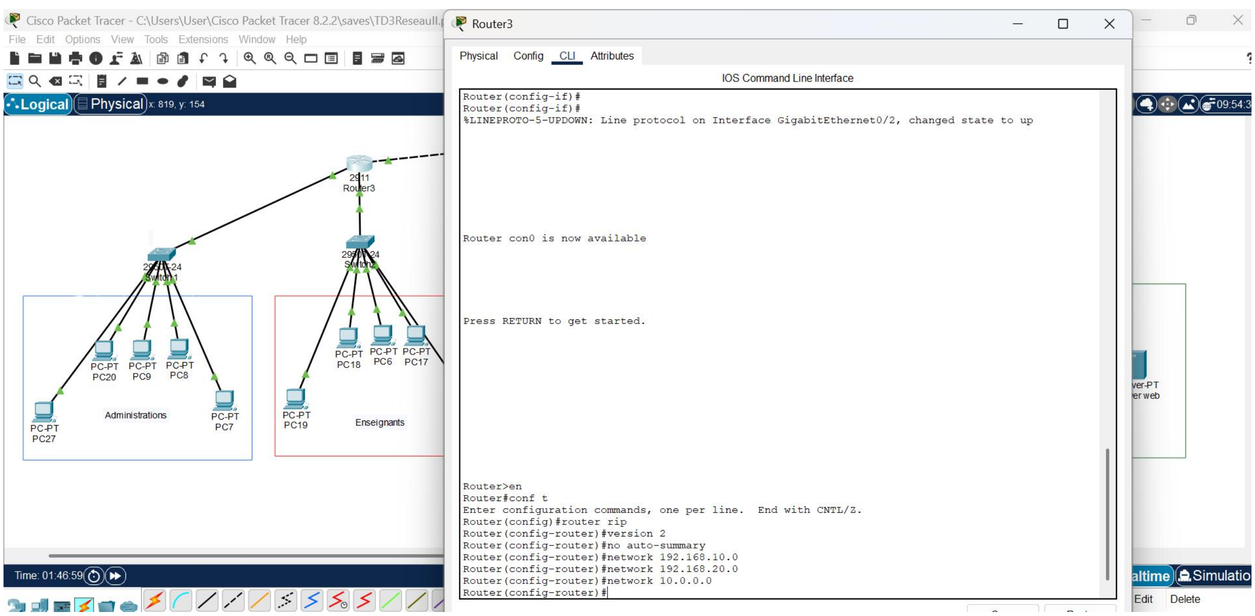


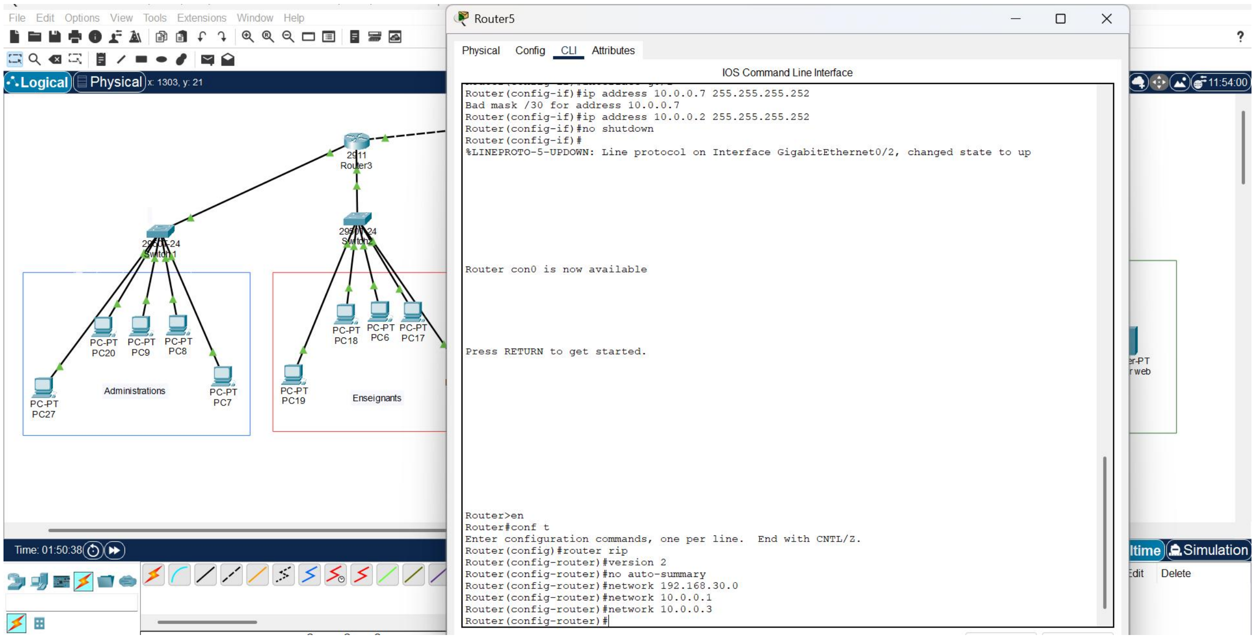

```
        --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.35.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.13.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.40.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface g0/2
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

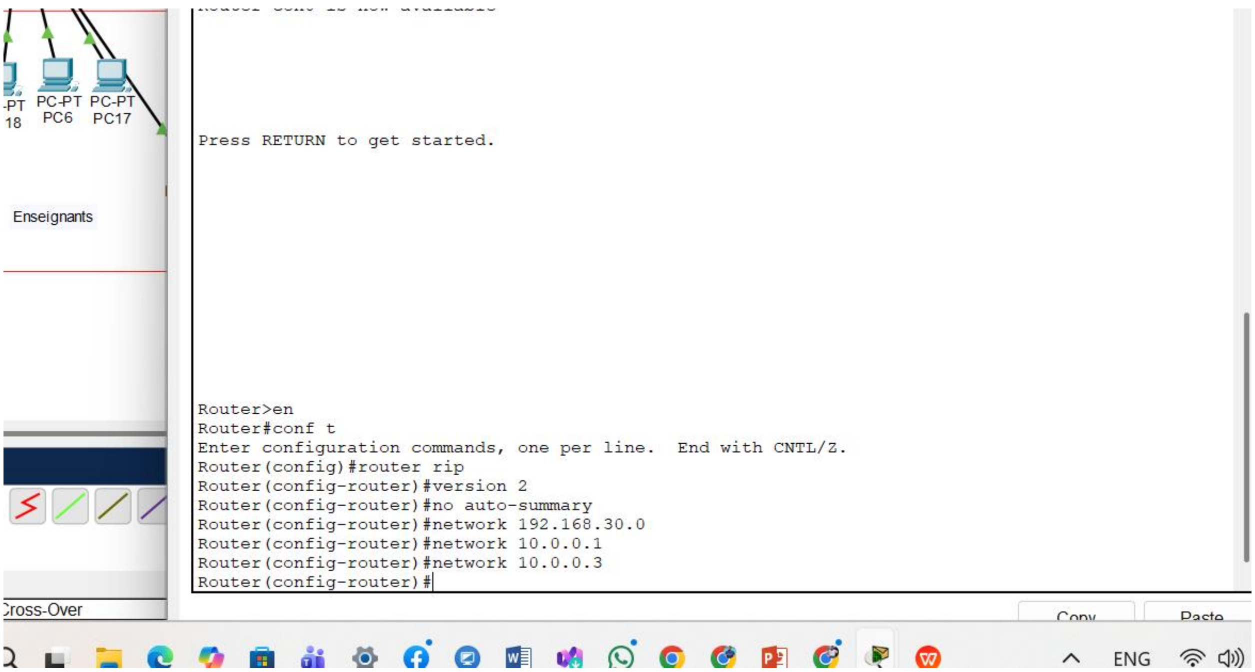
On peut voir toutes les connexions sont corrects

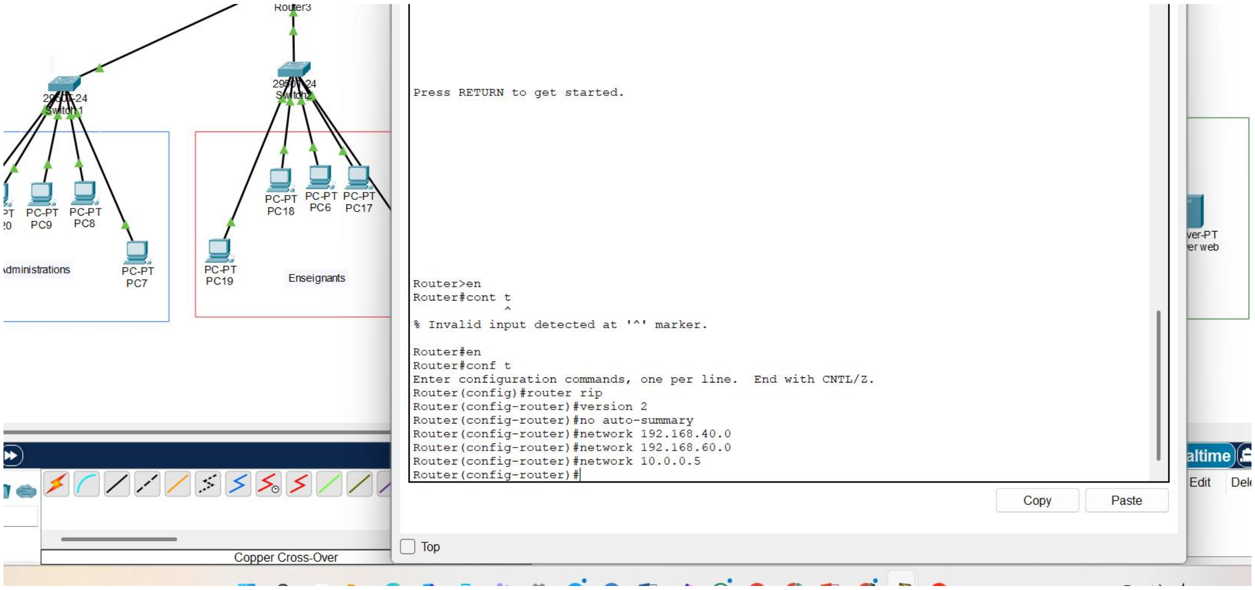


Configuration des routers avec RIP

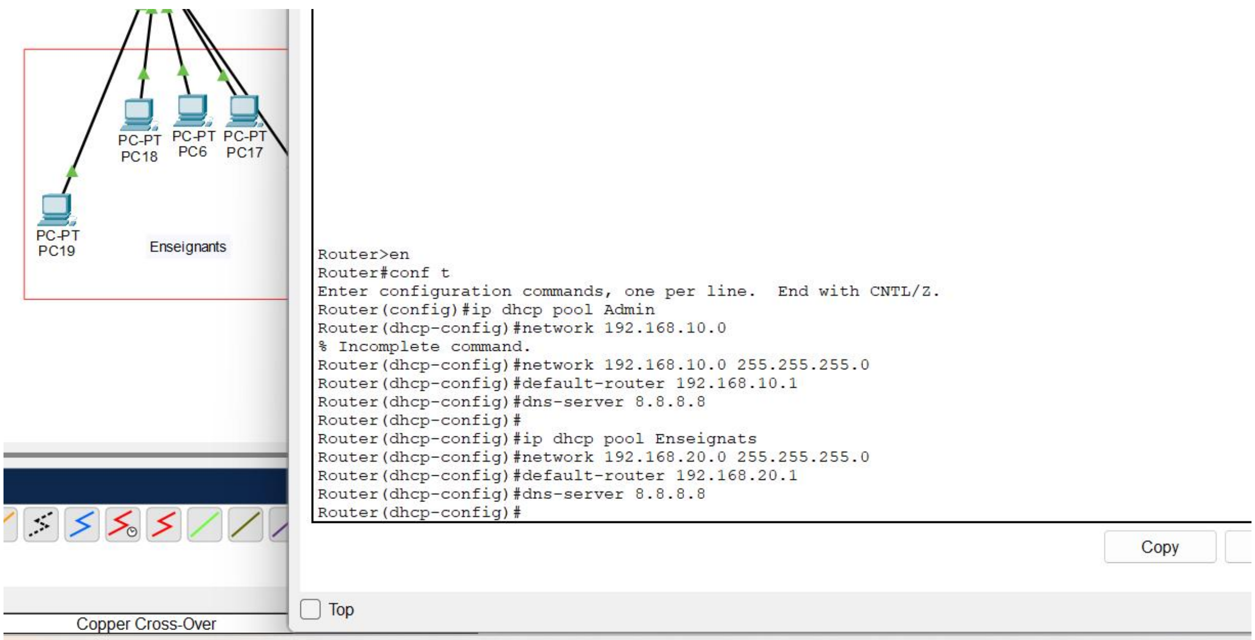


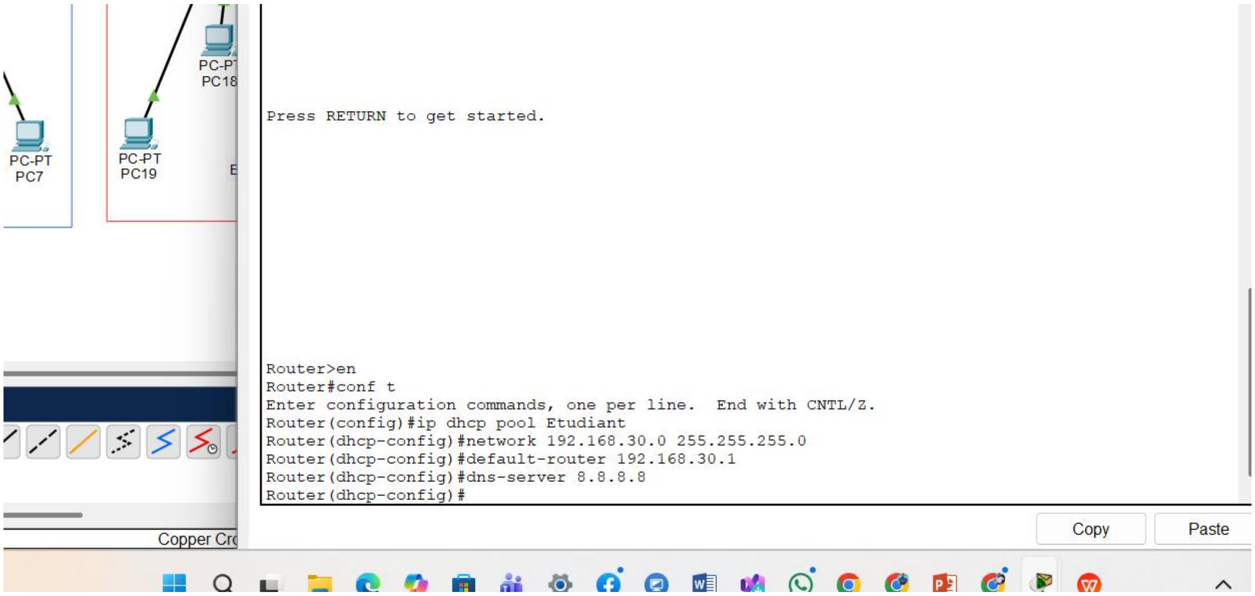

Router5
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#ip address 10.0.0.7 255.255.255.252
Bad mask /30 for address 10.0.0.7
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Router con0 is now available
Press RETURN to get started.
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.30.0
Router(config-router)#network 10.0.0.1
Router(config-router)#network 10.0.0.3
Router(config-router)#
Router3
Switch1
PC-PT PC20
PC-PT PC9
PC-PT PC8
PC-PT PC27
PC-PT PC7
Administrations
PC-PT PC18
PC-PT PC6
PC-PT PC17
PC-PT PC19
Enseignants
Time: 01:50:38
Simulation

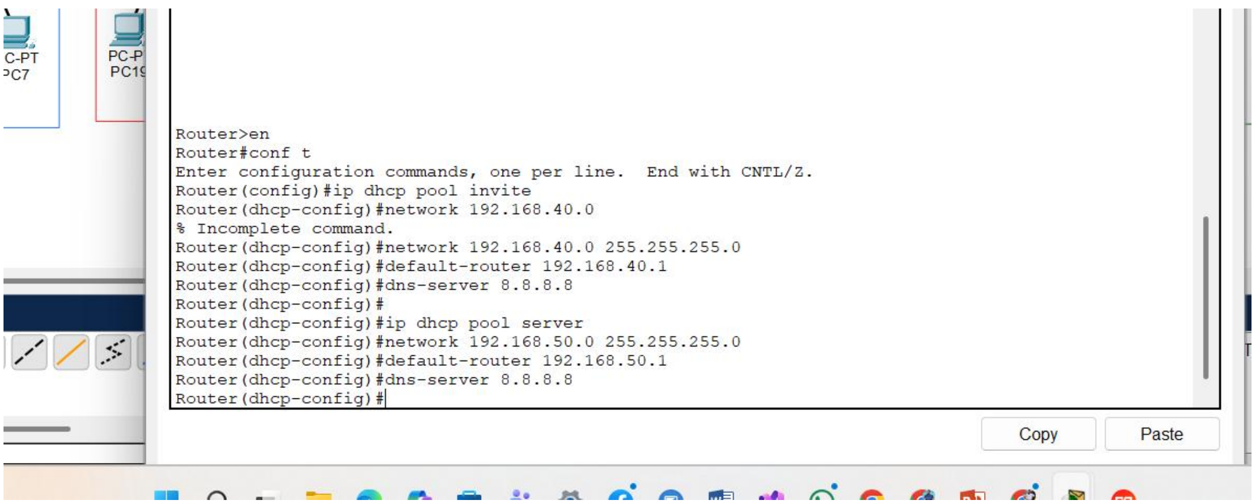

PC-PT PC6
PC-PT PC17
Enseignants
Press RETURN to get started.
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.30.0
Router(config-router)#network 10.0.0.1
Router(config-router)#network 10.0.0.3
Router(config-router)#
Cross-Over
ENG



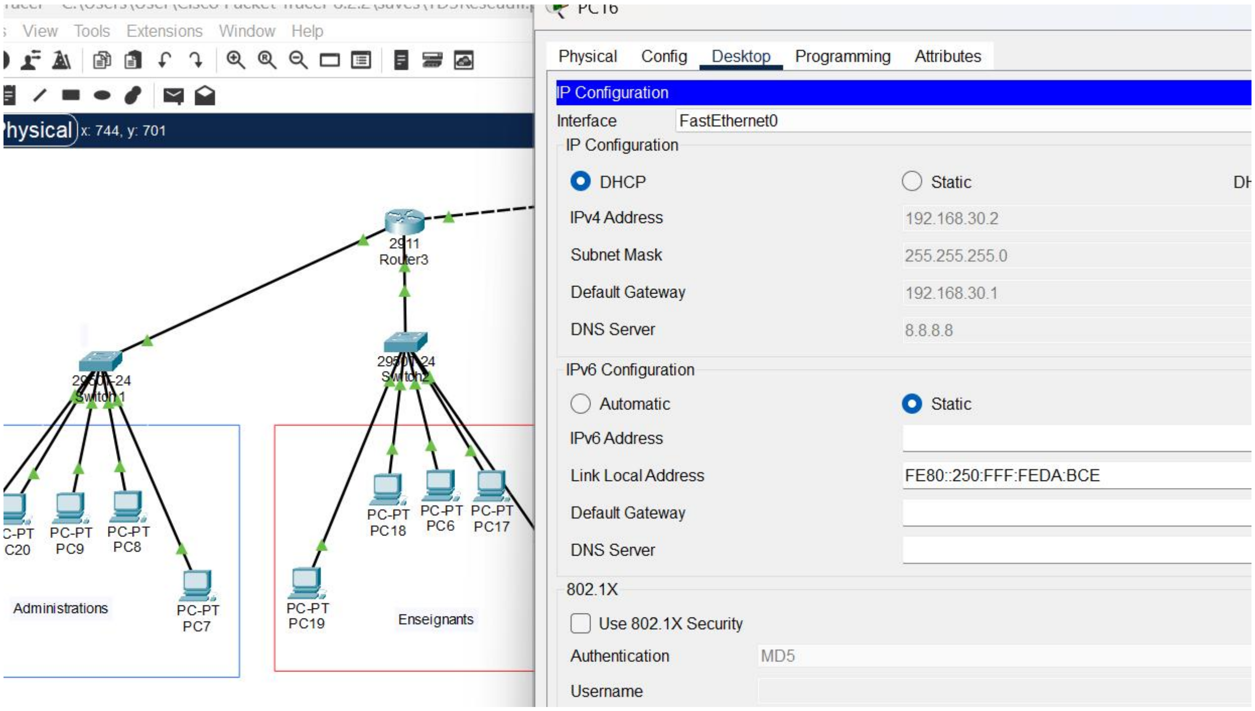
Configuration de DHCP avec les routeurs pour chaque VLANs

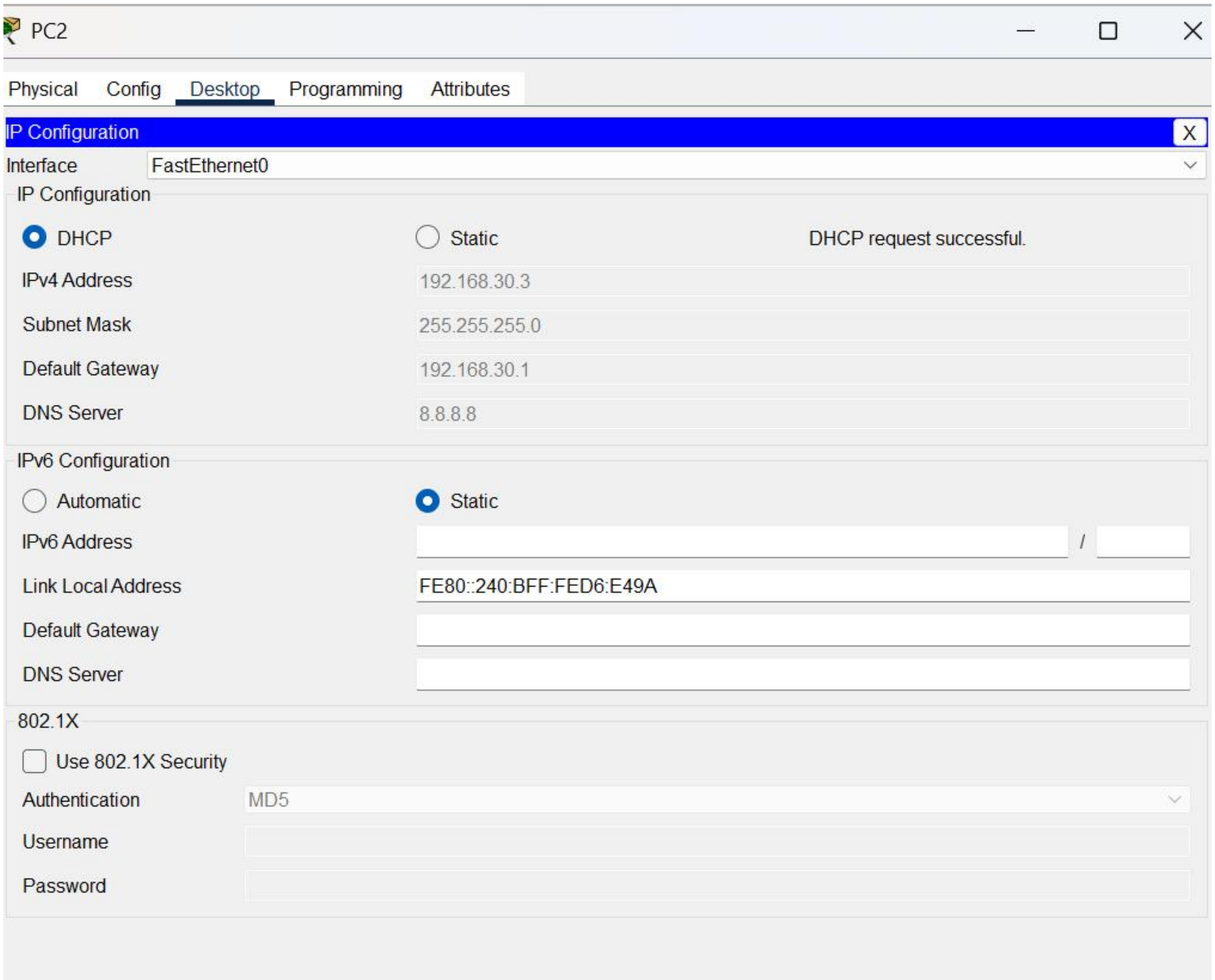



PC-PT
PC18
PC-PT
PC7
PC-PT
PC19
Press RETURN to get started.
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool Etudiant
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#
Copy
Paste

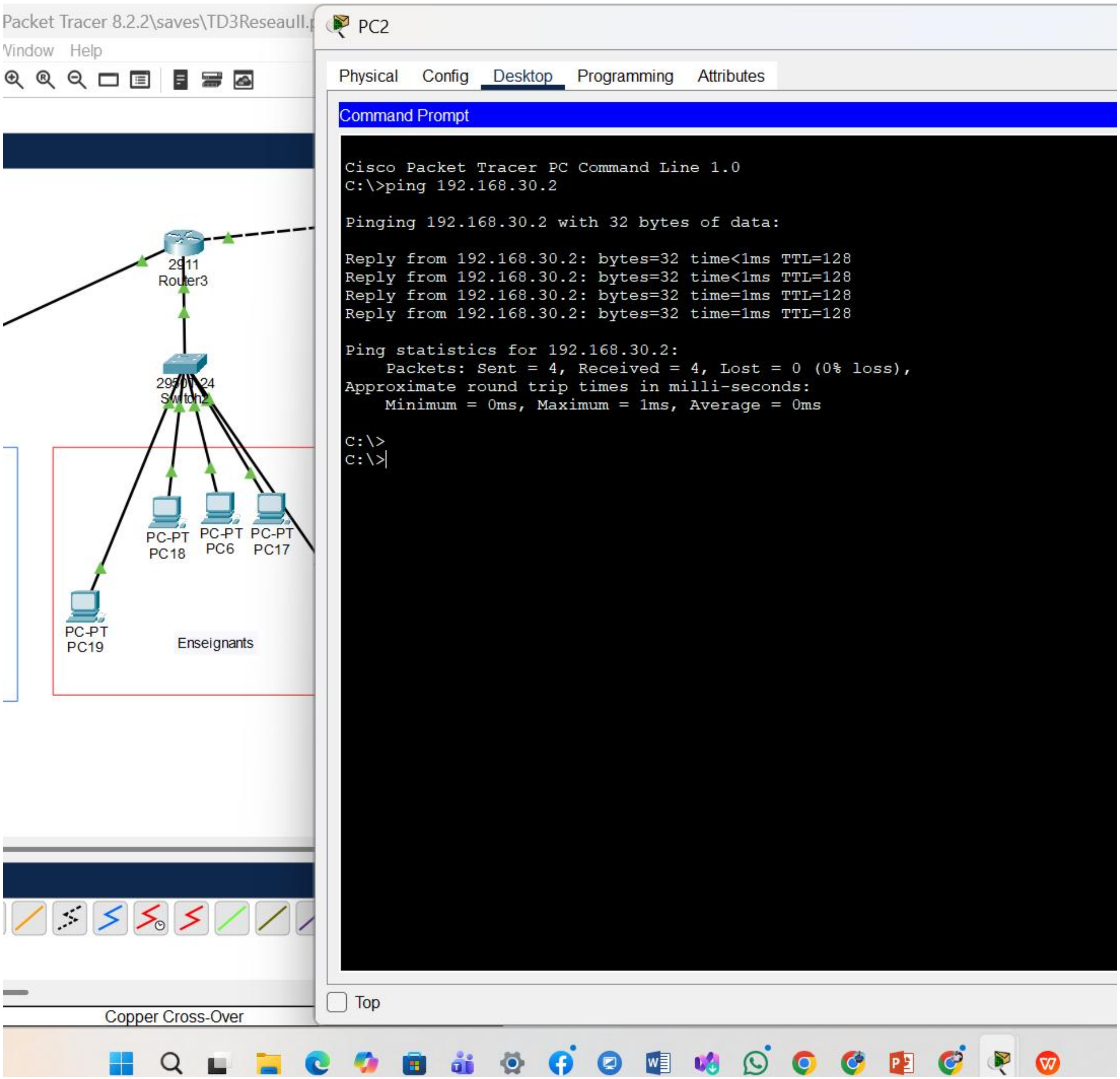

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool invite
Router(dhcp-config)#network 192.168.40.0
% Incomplete command.
Router(dhcp-config)#network 192.168.40.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.40.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool server
Router(dhcp-config)#network 192.168.50.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.50.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#
Copy
Paste

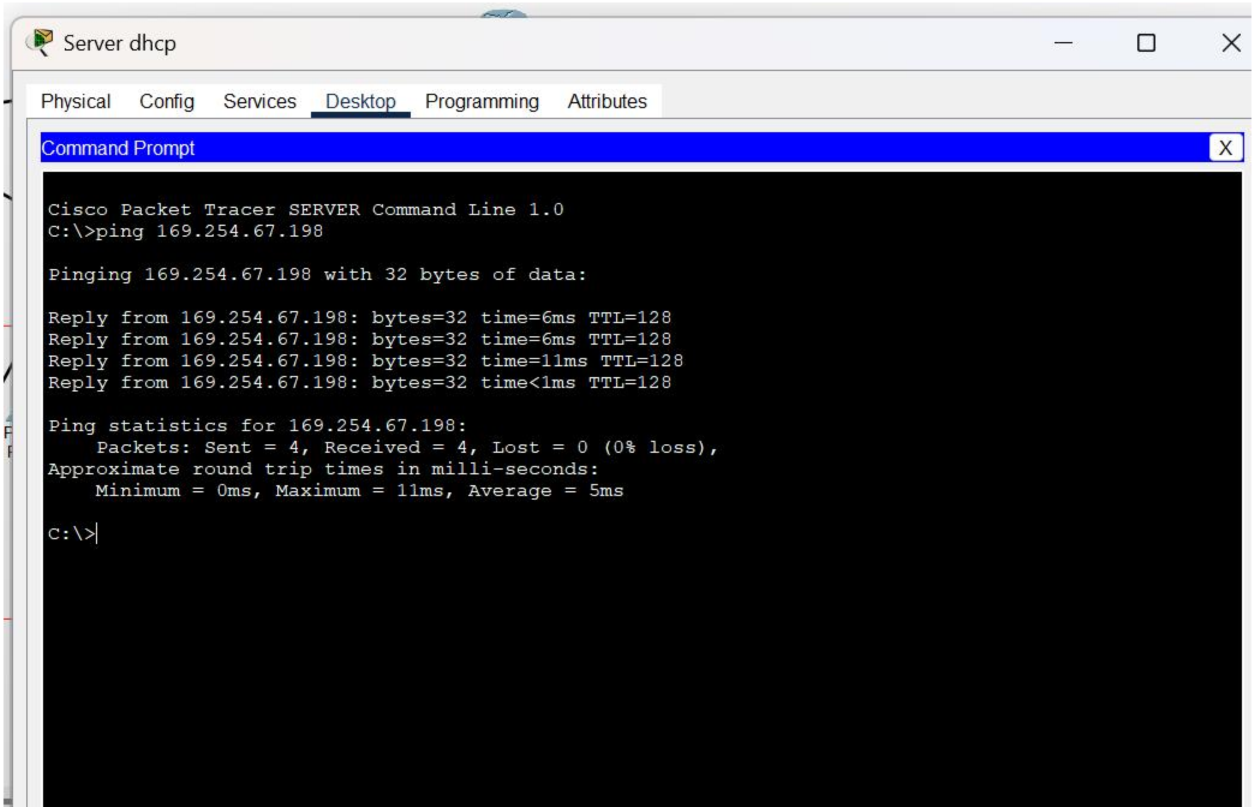
On peut voir le déploiement des address ip avec le DHCP attribuer aux PC



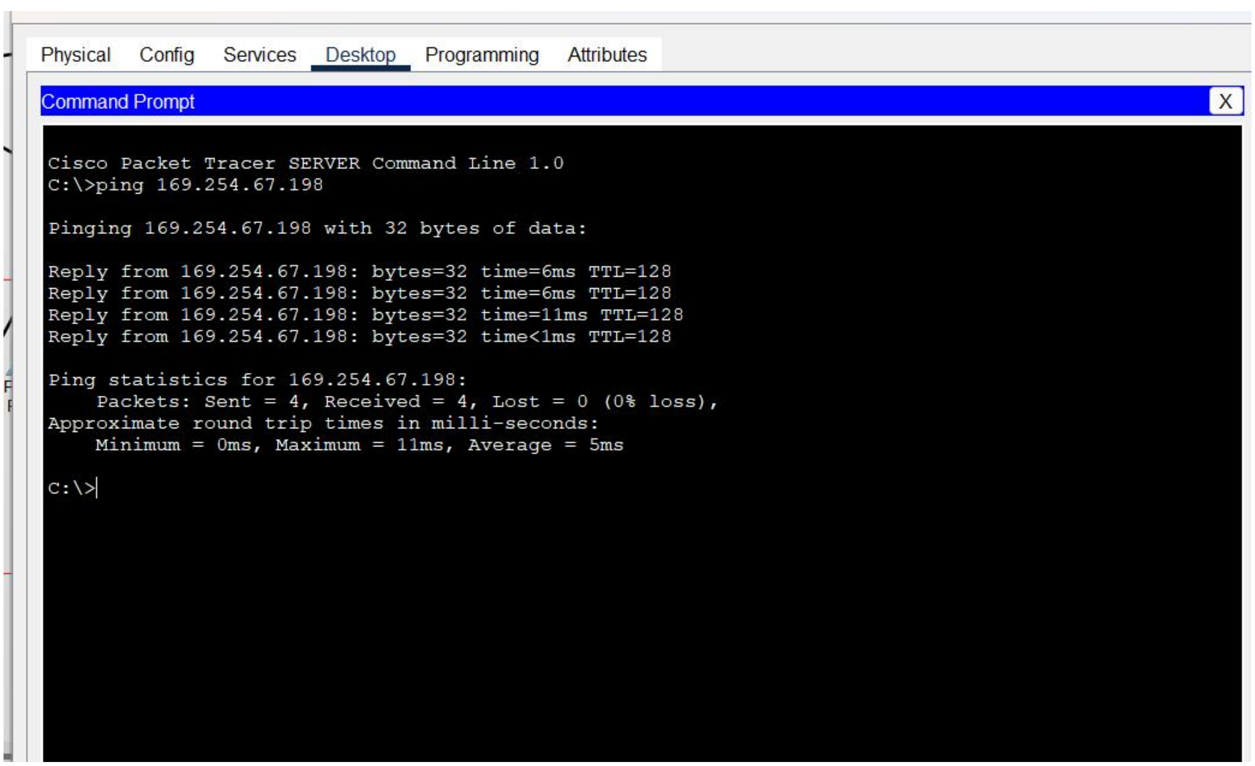

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.30.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::240:BFF:FED6:E49A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Ping entre les Pc





Ici je vais faire RIP avec la commande ping sur des PC de differents Vlan

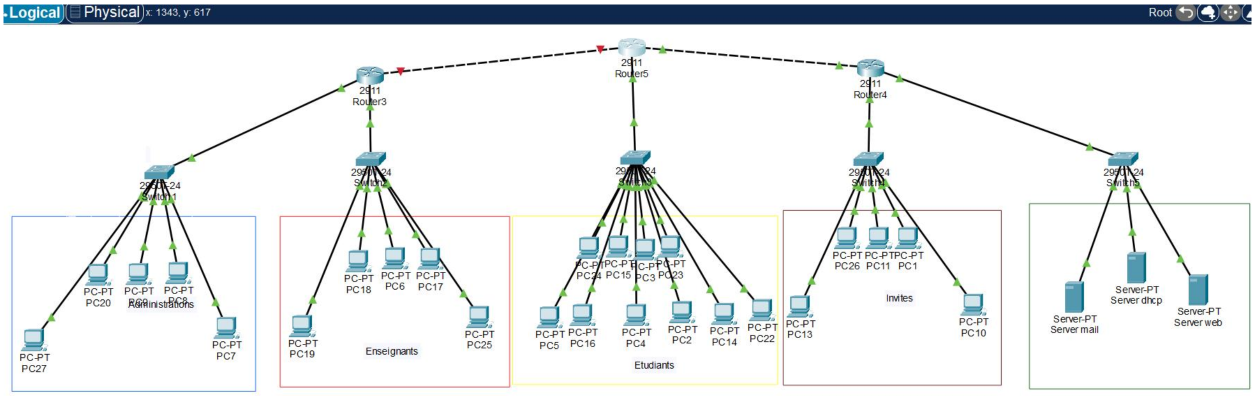


```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/1

Router>
```

Maintenant je vais couper un lien

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
```

On peut voir clairement ici qu’on a couper un lien




```
C:\>ping 192.169.25.4

Pinging 192.169.25.4 with 32 bytes of data:

Reply from 192.168.15.1: Destination host unreachable.
Request timed out.
Reply from 192.168.15.1: Destination host unreachable.
Reply from 192.168.15.1: Destination host unreachable.

Ping statistics for 192.169.25.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Apres on va établir la connexion tel quel était

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0

Router>
```

Copy

```
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.1 0.0.0.255 area 0
Router(config-router)#
```

Copper Cro

Configuration OSPF

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.15.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.2 0.0.0.255 area 0
Router(config-router)#network 192.168.25.0 0.0.0.255 area 0
Router(config-router)#
```

```
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.1 0.0.0.255 area 0
Router(config-router)#network 10.0.0.2 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.1 0.0.0.255 area 0
Router(config-router)#network 192.168.30.1 0.0.0.255 area 0
Router(config-router)#network 10.0.0.2 0.0.0.252 area 0
OSPF: Invalid address/mask combination (discontiguous mask)
Router(config-router)#network 10.0.0.2 0.0.0.3 area 0
Router(config-router)#
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 192.168.35.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.5 0.0.0.3 area 0
Router(config-router)#
```

Test OSPF

```
Router1>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address
Interface
192.168.20.1      1   FULL/DR         00:00:39    10.0.0.1
GigabitEthernet0/1
192.168.50.1      1   FULL/DR         00:00:39    10.0.0.6
GigabitEthernet0/2
Router1>
```

Ctrl+F6 to exit CLI focus   Copy   Paste

Je fais ici une verification

```
%SYS-5-CONFIG_I: Configured from console by console

Router2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/1
L       192.168.50.1/32 is directly connected, GigabitEthernet0/1
```

Configuration EIGRP

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/
Router(config)#router eigrp 100
Router(config-router)#network 192.168.15.0
Router(config-router)#network 192.168.25.0
Router(config-router)#network 10.0.0.2
Router(config-router)#no auto-summary
Router(config-router)#
Router(config-router)#
```

Top

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/
Router(config)#router eight 100
                          ^
% Invalid input detected at '^' marker.

Router(config)#router eigrp 100
Router(config-router)#network 192.168.35.0
Router(config-router)#network 192.168.40.0
Router(config-router)#network 10.0.0.5
Router(config-router)#no auto-summary
Router(config-router)#
```

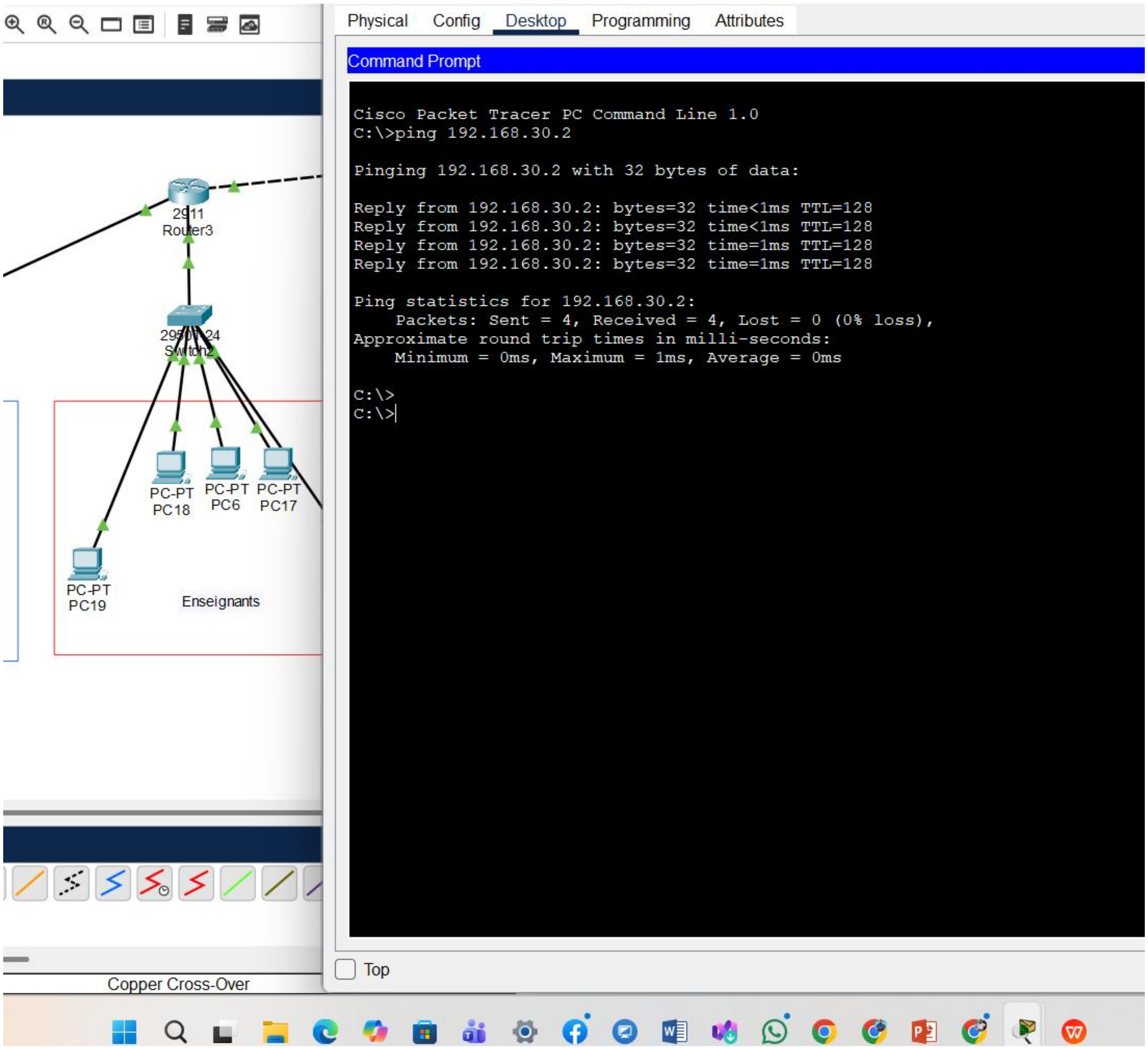
```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 100
Router(config-router)#network 192.168.30.0
Router(config-router)#network 192.168.40.0
Router(config-router)#network 10.0.0.2
Router(config-router)#no auto-summary
Router(config-router)#
```

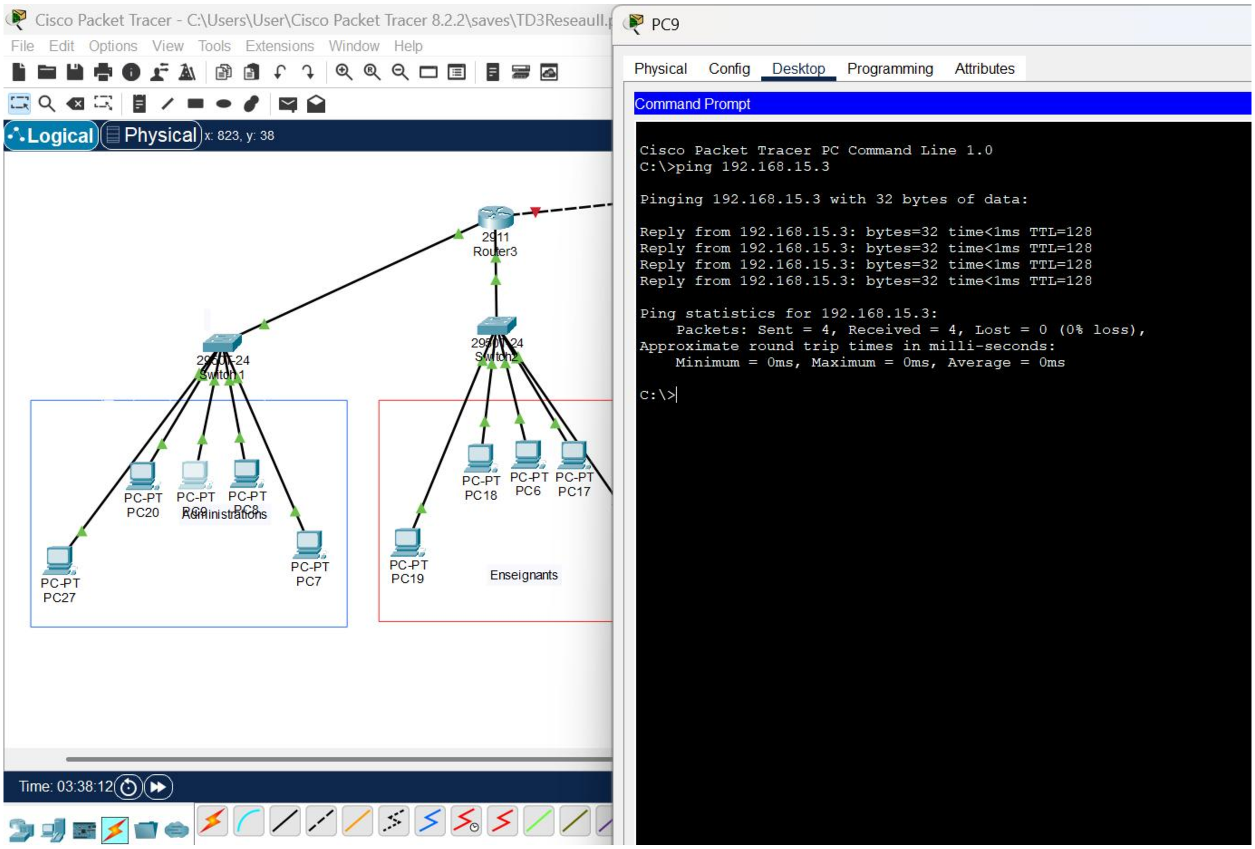
Teste EIGRP

```
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)         Cnt Num
0   10.0.0.2        Gig0/2          11   00:02:42  40     1000  0   9

Router0#
```

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0

Router#
```

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter a
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
     192.168.35.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.35.0/24 is directly connected, GigabitEthernet0/0
L       192.168.35.1/32 is directly connected, GigabitEthernet0/0
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/1

Router#
```


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2
Pinging 192.168.30.2 with 32 bytes of data:
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time=1ms TTL=128
Reply from 192.168.30.2: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.30.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
C:\>
Top
2911
Router3
Switch2
PC-PT
PC18
PC-PT
PC6
PC-PT
PC17
PC-PT
PC19
Enseignants
Copper Cross-Over



Questions ? Réponses

1. Pourquoi OSPF est plus efficace que RIP ?

- **RIP** utilise un algorithme simple basé sur la distance (nombre de sauts), limité à 15 sauts.
- **OSPF** (Open Shortest Path First) utilise l'algorithme de Dijkstra et calcule le chemin le plus court en fonction de plusieurs métriques (bande passante, coût).
- OSPF converge plus rapidement, supporte de grands réseaux et évite les boucles plus efficacement.

En résumé : OSPF est plus adapté aux réseaux complexes et de grande taille.

2. Quelle différence entre protocole classful et classless ?

- **Classful** : les anciens protocoles (comme RIP v1) ne transmettent pas l'information sur le masque de sous-réseau. Ils supposent des classes A, B ou C fixes.
- **Classless** : les protocoles modernes (RIP v2, OSPF, EIGRP) incluent le masque de sous-réseau dans leurs annonces. Cela permet le **VLSM (Variable Length Subnet Masking)** et une meilleure utilisation des adresses IP.

En résumé : classless = plus flexible et efficace dans l’adressage.

3. Pourquoi EIGRP est plus rapide ?

- **EIGRP** (Enhanced Interior Gateway Routing Protocol) utilise un algorithme propriétaire Cisco appelé **DUAL** (Diffusing Update Algorithm).
- Il calcule les routes de secours en avance, ce qui permet une **convergence quasi instantanée** en cas de panne.
- Il combine les avantages de distance vector et link state, avec des métriques multiples (bande passante, délai, fiabilité, charge).

  En résumé : EIGRP est plus rapide car il anticipe les changements et converge très vite.

4. Quel protocole choisir pour une grande entreprise et pourquoi ?

- Pour une grande entreprise, le choix se fait généralement entre **OSPF** et **EIGRP**.
- **OSPF** est standard, ouvert et interopérable entre différents constructeurs (Cisco, Juniper, etc.).
- **EIGRP** est très performant mais longtemps resté propriétaire Cisco (même si aujourd’hui il est partiellement ouvert).

  En pratique : **OSPF** est le protocole recommandé pour une grande entreprise car il est robuste, évolutif et compatible multi- constructeurs.